ISLA LP BENCHMARK

Reporte de Rendimiento

Problemas evaluados: 150

Solvers comparados: 5

Tasa de exito: 100.0%

Generado: 03 de May de 2026 a las 23:16

RESUMEN ESTADISTICO

Metricas Globales:

Total de pruebas	Exitosas	Fallidas	Tasa de exito
150	150	0	100.0%

Ranking por Velocidad (tiempo promedio):

1. highs — 124.87 ms (Mas rapido)
2. scip — 126.01 ms
3. gurobi — 129.01 ms
4. glpk — 129.97 ms
5. cbc — 193.75 ms (Mas lento)

Tabla Comparativa:

Solver	Pruebas	Exito	Tiempo prom.	Tiempo total	Memoria peak
gurobi	30	100%	129.01ms	3870.45ms	145.8 MB
cbc	30	100%	193.75ms	5812.51ms	145.9 MB
scip	30	100%	126.01ms	3780.44ms	145.9 MB
highs	30	100%	124.87ms	3745.99ms	146.0 MB
glpk	30	100%	129.97ms	3899.01ms	146.0 MB

COMPARACION DETALLADA POR SOLVER

GUROBI

Ejecuciones:	30
Exitosas:	30 (100.0%)
Tiempo promedio:	129.015 ms
Tiempo minimo:	93.833 ms
Tiempo maximo:	199.494 ms
Desv. estandar:	27.452 ms
Memoria promedio:	0.22 MB
Memoria peak:	145.84 MB

CBC

Ejecuciones:	30
Exitosas:	30 (100.0%)
Tiempo promedio:	193.750 ms
Tiempo minimo:	149.032 ms
Tiempo maximo:	292.411 ms
Desv. estandar:	30.341 ms
Memoria promedio:	0.01 MB
Memoria peak:	145.88 MB

SCIP

Ejecuciones:	30
Exitosas:	30 (100.0%)
Tiempo promedio:	126.015 ms
Tiempo minimo:	105.847 ms
Tiempo maximo:	168.995 ms
Desv. estandar:	13.847 ms
Memoria promedio:	0.27 MB
Memoria peak:	145.93 MB

HIGHS

Ejecuciones:	30
Exitosas:	30 (100.0%)

Tiempo promedio:	124.866 ms
Tiempo minimo:	98.406 ms
Tiempo maximo:	162.881 ms
Desv. estandar:	17.382 ms
Memoria promedio:	0.11 MB
Memoria peak:	145.96 MB

GLPK

Ejecuciones:	30
Exitosas:	30 (100.0%)
Tiempo promedio:	129.967 ms
Tiempo minimo:	97.470 ms
Tiempo maximo:	239.113 ms
Desv. estandar:	30.834 ms
Memoria promedio:	0.01 MB
Memoria peak:	145.99 MB

RESULTADOS DETALLADOS

Problema: Problema_1

Solver	Estado	Valor Obj	Tiempo	Memoria	Iter	Nodos
gurobi	OPTIMAL	8.00	116.56ms	5.8MB	1	-
gurobi	OPTIMAL	8.00	102.45ms	0.0MB	1	-
gurobi	OPTIMAL	8.00	96.08ms	-	1	-
cbc	OPTIMAL	8.00	171.82ms	0.1MB	-	-
cbc	OPTIMAL	8.00	173.84ms	-	-	-
cbc	OPTIMAL	8.00	189.51ms	0.0MB	-	-
scip	OPTIMAL	8.00	153.66ms	6.3MB	-	-
scip	OPTIMAL	8.00	116.86ms	-	-	-
scip	OPTIMAL	8.00	110.07ms	0.1MB	-	-
highs	OPTIMAL	8.00	110.39ms	3.2MB	-	-
highs	OPTIMAL	8.00	116.40ms	-	-	-
highs	OPTIMAL	8.00	112.11ms	-	-	-
glpk	OPTIMAL	-22.00	120.89ms	0.2MB	-	-
glpk	OPTIMAL	-22.00	133.48ms	-	-	-
glpk	OPTIMAL	-22.00	135.67ms	-	-	-

Problema: Problema_10

Solver	Estado	Valor Obj	Tiempo	Memoria	Iter	Nodos
gurobi	OPTIMAL	7790.00	120.34ms	0.0MB	10	-
gurobi	OPTIMAL	7790.00	128.41ms	0.0MB	10	-
gurobi	OPTIMAL	7790.00	199.49ms	0.0MB	10	-
cbc	OPTIMAL	7790.00	193.05ms	0.0MB	-	-
cbc	OPTIMAL	7790.00	174.93ms	0.0MB	-	-
cbc	OPTIMAL	7790.00	175.53ms	-	-	-
scip	OPTIMAL	7790.00	124.39ms	0.0MB	-	-
scip	OPTIMAL	7790.00	131.93ms	0.0MB	-	-
scip	OPTIMAL	7790.00	139.09ms	-	-	-
highs	OPTIMAL	7790.00	127.81ms	0.0MB	-	-
highs	OPTIMAL	7790.00	118.15ms	-	-	-
highs	OPTIMAL	7790.00	102.88ms	-	-	-
glpk	OPTIMAL	7790.00	109.31ms	-	-	-
glpk	OPTIMAL	7790.00	128.04ms	-	-	-
glpk	OPTIMAL	7790.00	115.50ms	-	-	-

Problema: Problema_2

Solver	Estado	Valor Obj	Tiempo	Memoria	Iter	Nodos
gurobi	OPTIMAL	406.67	111.84ms	0.4MB	3	-
gurobi	OPTIMAL	406.67	106.40ms	0.0MB	3	-
gurobi	OPTIMAL	406.67	106.33ms	0.0MB	3	-
cbc	OPTIMAL	406.67	149.03ms	0.0MB	-	-
cbc	OPTIMAL	406.67	175.62ms	-	-	-
cbc	OPTIMAL	406.67	199.14ms	0.0MB	-	-
scip	OPTIMAL	406.67	169.00ms	0.8MB	-	-
scip	OPTIMAL	406.67	135.96ms	0.4MB	-	-
scip	OPTIMAL	406.67	109.35ms	-	-	-
highs	OPTIMAL	406.67	98.41ms	0.1MB	-	-
highs	OPTIMAL	406.67	102.32ms	-	-	-
highs	OPTIMAL	406.67	104.73ms	-	-	-
glpk	OPTIMAL	-	108.88ms	-	-	-
glpk	OPTIMAL	-	131.12ms	-	-	-
glpk	OPTIMAL	-	139.41ms	-	-	-

Problema: Problema_3

gurobi	OPTIMAL	68.00	166.55ms	0.1MB	3	-
gurobi	OPTIMAL	68.00	93.83ms	0.0MB	3	-
gurobi	OPTIMAL	68.00	195.67ms	0.0MB	3	-
cbc	OPTIMAL	68.00	167.72ms	0.0MB	-	-
cbc	OPTIMAL	68.00	192.68ms	0.0MB	-	-
cbc	OPTIMAL	68.00	232.44ms	-	-	-
scip	OPTIMAL	68.00	141.80ms	0.2MB	-	-
scip	OPTIMAL	68.00	115.11ms	0.0MB	-	-
scip	OPTIMAL	68.00	140.62ms	0.0MB	-	-
highs	OPTIMAL	68.00	115.56ms	-	-	-
highs	OPTIMAL	68.00	129.24ms	-	-	-
highs	OPTIMAL	68.00	152.41ms	0.0MB	-	-
glpk	OPTIMAL	68.00	222.32ms	-	-	-
glpk	OPTIMAL	68.00	149.76ms	-	-	-
glpk	OPTIMAL	68.00	132.63ms	-	-	-

Problema: Problema_4

Solver	Estado	Valor Obj	Tiempo	Memoria	Iter	Nodos
gurobi	OPTIMAL	1253.33	106.12ms	0.0MB	9	-
gurobi	OPTIMAL	1253.33	111.01ms	0.0MB	9	-
gurobi	OPTIMAL	1253.33	120.09ms	0.0MB	9	-
cbc	OPTIMAL	1253.33	180.61ms	0.0MB	-	-
cbc	OPTIMAL	1253.33	213.12ms	0.0MB	-	-
cbc	OPTIMAL	1253.33	204.75ms	0.0MB	-	-
scip	OPTIMAL	1253.33	108.44ms	0.0MB	-	-
scip	OPTIMAL	1253.33	105.85ms	0.0MB	-	-
scip	OPTIMAL	1253.33	111.24ms	0.0MB	-	-
highs	OPTIMAL	1253.33	105.84ms	-	-	-
highs	OPTIMAL	1253.33	125.39ms	-	-	-
highs	OPTIMAL	1253.33	142.99ms	-	-	-
glpk	OPTIMAL	-	142.62ms	-	-	-
glpk	OPTIMAL	-	131.08ms	0.0MB	-	-
glpk	OPTIMAL	-	103.34ms	-	-	-

Problema: Problema_5

Solver	Estado	Valor Obj	Tiempo	Memoria	Iter	Nodos
gurobi	OPTIMAL	80.00	116.30ms	0.0MB	2	-
gurobi	OPTIMAL	80.00	115.15ms	0.0MB	2	-
gurobi	OPTIMAL	80.00	124.35ms	0.0MB	2	-
cbc	OPTIMAL	80.00	176.37ms	0.0MB	-	-
cbc	OPTIMAL	80.00	215.33ms	-	-	-
cbc	OPTIMAL	80.00	225.66ms	-	-	-
scip	OPTIMAL	80.00	117.45ms	0.0MB	-	-
scip	OPTIMAL	80.00	108.59ms	-	-	-
scip	OPTIMAL	80.00	129.32ms	-	-	-
highs	OPTIMAL	80.00	121.04ms	0.0MB	-	-
highs	OPTIMAL	80.00	115.36ms	-	-	-
highs	OPTIMAL	80.00	117.43ms	-	-	-
glpk	OPTIMAL	80.00	144.51ms	-	-	-
glpk	OPTIMAL	80.00	153.05ms	-	-	-
glpk	OPTIMAL	80.00	100.98ms	-	-	-

Problema: Problema_6

Solver	Estado	Valor Obj	Tiempo	Memoria	Iter	Nodos
gurobi	OPTIMAL	18233.33	184.95ms	0.1MB	3	-
gurobi	OPTIMAL	18233.33	118.40ms	0.0MB	3	-
gurobi	OPTIMAL	18233.33	118.98ms	0.0MB	3	-
cbc	OPTIMAL	18233.33	175.97ms	0.0MB	-	-
cbc	OPTIMAL	18233.33	292.41ms	0.0MB	-	-
cbc	OPTIMAL	18233.33	198.62ms	-	-	-
scip	OPTIMAL	18233.33	122.10ms	0.0MB	-	-

scip	OPTIMAL	18233.33	125.28ms	-	-	-
scip	OPTIMAL	18233.33	132.93ms	-	-	-
highs	OPTIMAL	18233.33	123.28ms	0.0MB	-	-
highs	OPTIMAL	18233.33	141.31ms	-	-	-
highs	OPTIMAL	18233.33	162.88ms	-	-	-
glpk	OPTIMAL	-	126.76ms	-	-	-
glpk	OPTIMAL	-	119.16ms	-	-	-
glpk	OPTIMAL	-	109.76ms	0.0MB	-	-

Problema: Problema_7

Solver	Estado	Valor Obj	Tiempo	Memoria	Iter	Nodos
gurobi	OPTIMAL	403.08	126.00ms	0.0MB	4	-
gurobi	OPTIMAL	403.08	165.63ms	0.0MB	4	-
gurobi	OPTIMAL	403.08	129.03ms	-	4	-
cbc	OPTIMAL	403.08	260.04ms	0.0MB	-	-
cbc	OPTIMAL	403.08	187.19ms	-	-	-
cbc	OPTIMAL	403.08	156.41ms	-	-	-
scip	OPTIMAL	403.08	125.94ms	0.0MB	-	-
scip	OPTIMAL	403.08	134.23ms	-	-	-
scip	OPTIMAL	403.08	125.62ms	-	-	-
highs	OPTIMAL	403.08	121.50ms	-	-	-
highs	OPTIMAL	403.08	148.95ms	-	-	-
highs	OPTIMAL	403.08	121.22ms	0.0MB	-	-
glpk	OPTIMAL	403.08	126.80ms	-	-	-
glpk	OPTIMAL	403.08	100.05ms	0.0MB	-	-
glpk	OPTIMAL	403.08	107.67ms	-	-	-

Problema: Problema_8

Solver	Estado	Valor Obj	Tiempo	Memoria	Iter	Nodos
gurobi	OPTIMAL	1020.00	128.31ms	0.0MB	4	-
gurobi	OPTIMAL	1020.00	116.70ms	-	4	-
gurobi	OPTIMAL	1020.00	119.62ms	-	4	-
cbc	OPTIMAL	1020.00	233.53ms	0.0MB	-	-
cbc	OPTIMAL	1020.00	181.38ms	-	-	-
cbc	OPTIMAL	1020.00	159.20ms	-	-	-
scip	OPTIMAL	1020.00	126.06ms	-	-	-
scip	OPTIMAL	1020.00	119.31ms	-	-	-
scip	OPTIMAL	1020.00	124.31ms	-	-	-
highs	OPTIMAL	1020.00	144.17ms	-	-	-
highs	OPTIMAL	1020.00	157.79ms	-	-	-
highs	OPTIMAL	1020.00	135.42ms	-	-	-
glpk	OPTIMAL	-	131.32ms	-	-	-
glpk	OPTIMAL	-	239.11ms	-	-	-
glpk	OPTIMAL	-	117.64ms	-	-	-

Problema: Problema_9

Solver	Estado	Valor Obj	Tiempo	Memoria	Iter	Nodos
gurobi	OPTIMAL	71730.77	131.94ms	0.0MB	8	-
gurobi	OPTIMAL	71730.77	134.29ms	0.0MB	8	-
gurobi	OPTIMAL	71730.77	159.62ms	-	8	-
cbc	OPTIMAL	71730.77	187.05ms	0.0MB	-	-
cbc	OPTIMAL	71730.77	182.81ms	0.0MB	-	-
cbc	OPTIMAL	71730.77	186.76ms	0.0MB	-	-
scip	OPTIMAL	71730.77	119.94ms	-	-	-
scip	OPTIMAL	71730.77	121.25ms	-	-	-
scip	OPTIMAL	71730.77	134.72ms	-	-	-
highs	OPTIMAL	71730.77	143.57ms	-	-	-
highs	OPTIMAL	71730.77	125.19ms	-	-	-
highs	OPTIMAL	71730.77	102.25ms	-	-	-
glpk	OPTIMAL	-	97.47ms	-	-	-
glpk	OPTIMAL	-	112.00ms	-	-	-
glpk	OPTIMAL	-	108.69ms	0.0MB	-	-

DEFINICIONES DE PROBLEMAS

Problema: Problema_1

```
MAX Z = -1.0x +4.0y
-3.0x +1.0y <= 6.0
1.0x +2.0y <= 4.0
y >= -3.0
```

Problema: Problema_2

```
MAX Z = 3.0x +5.0y +2.0z
1.0x +1.0y +1.0z <= 100.0
2.0x +1.0y <= 80.0
1.0y +3.0z <= 90.0
1.0x +1.0z <= 70.0
x >= 0.0
y >= 0.0
z >= 0.0
```

Problema: Problema_3

```
MIN Z = 5.0x +4.0y +3.0z +2.0w
1.0x +1.0y >= 10.0
1.0x +1.0z >= 8.0
1.0y +1.0w >= 12.0
2.0x +1.0y +1.0z <= 50.0
1.0x +3.0y +2.0w <= 60.0
3.0z +1.0w <= 40.0
1.0x +1.0y +1.0z +1.0w >= 20.0
x >= 0.0
y >= 0.0
z >= 0.0
w >= 0.0
```

Problema: Problema_4

```
MAX Z = 2.0x1 +3.0x2 +4.0x3 +5.0x4 +6.0x5
1.0x1 +1.0x2 +1.0x3 <= 200.0
1.0x3 +1.0x4 +1.0x5 <= 300.0
2.0x1 +1.0x2 <= 150.0
1.0x4 +3.0x5 <= 250.0
1.0x1 +1.0x5 <= 100.0
1.0x2 +1.0x3 +1.0x4 <= 180.0
x1 >= 0.0
x2 >= 0.0
x3 >= 0.0
x4 >= 0.0
x5 >= 0.0
```

Problema: Problema_5

```
MIN Z = 5.0x1 +8.0x2 +3.0x3 +6.0x4
1.0x1 +1.0x2 >= 10.0
1.0x2 +1.0x3 >= 8.0
1.0x3 +1.0x4 >= 6.0
1.0x1 +1.0x4 >= 8.0
2.0x1 +1.0x2 +1.0x3 <= 30.0
1.0x2 +2.0x3 +1.0x4 <= 35.0
```

```
3.0x1 +1.0x3 +1.0x4 <= 40.0
1.0x1 +1.0x2 +1.0x3 +1.0x4 >= 20.0
1.0x1 +2.0x2 +3.0x3 +1.0x4 <= 50.0
x1 >= 0.0
x2 >= 0.0
```

Problema: Problema_6

```
MAX Z = 10.0x1 +15.0x2 +20.0x3 +12.0x4 +18.0x5 +25.0x6
1.0x1 +1.0x2 +1.0x3 +1.0x4 +1.0x5 +1.0x6 <= 1000.0
2.0x1 +1.0x2 +3.0x3 +1.0x4 +2.0x5 +4.0x6 <= 2000.0
1.0x1 +2.0x2 +1.0x3 +3.0x4 +1.0x5 +2.0x6 <= 1800.0
3.0x1 +1.0x2 +2.0x3 +1.0x4 +3.0x5 +1.0x6 <= 2200.0
1.0x1 +1.0x3 +1.0x5 <= 600.0
1.0x2 +1.0x4 +1.0x6 <= 700.0
x1 >= 0.0
x2 >= 0.0
x3 >= 0.0
x4 >= 0.0
x5 >= 0.0
```

Problema: Problema_7

```
MIN Z = 7.0x1 +9.0x2 +6.0x3 +8.0x4 +10.0x5
2.0x1 +1.0x2 +3.0x3 +1.0x4 +2.0x5 >= 100.0
1.0x1 +3.0x2 +1.0x3 +2.0x4 +1.0x5 >= 80.0
1.0x2 +2.0x3 +3.0x4 +1.0x5 >= 90.0
3.0x1 +1.0x2 +1.0x3 +2.0x4 +3.0x5 >= 120.0
1.0x1 +1.0x2 +1.0x3 +1.0x4 +1.0x5 <= 200.0
2.0x1 +3.0x2 +1.0x3 +1.0x4 +1.0x5 >= 70.0
1.0x1 +2.0x2 +3.0x3 +2.0x4 +1.0x5 <= 180.0
x1 >= 0.0
x2 >= 0.0
x3 >= 0.0
x4 >= 0.0
```

Problema: Problema_8

```
MAX Z = 12.0x +15.0y +10.0z +8.0w
1.0x +1.0y <= 50.0
1.0y +1.0z <= 60.0
1.0z +1.0w <= 70.0
1.0x +1.0w <= 40.0
2.0x +1.0y +1.0z <= 100.0
1.0x +2.0y +1.0z +1.0w <= 120.0
1.0x +1.0y +2.0z +1.0w <= 110.0
1.0x +1.0y +1.0z +2.0w <= 90.0
x >= 0.0
y >= 0.0
z >= 0.0
```

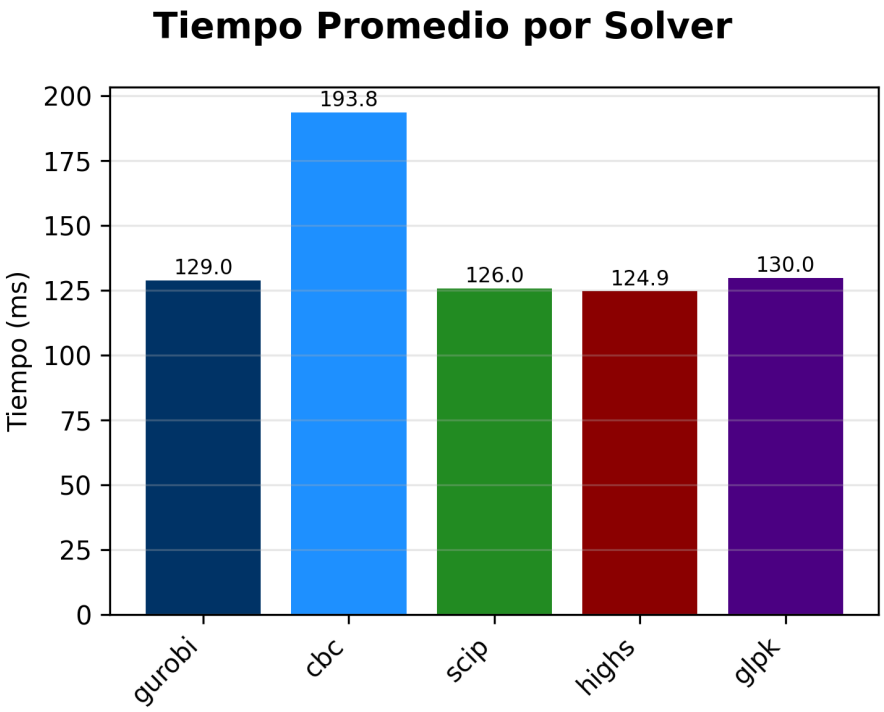
Problema: Problema_9

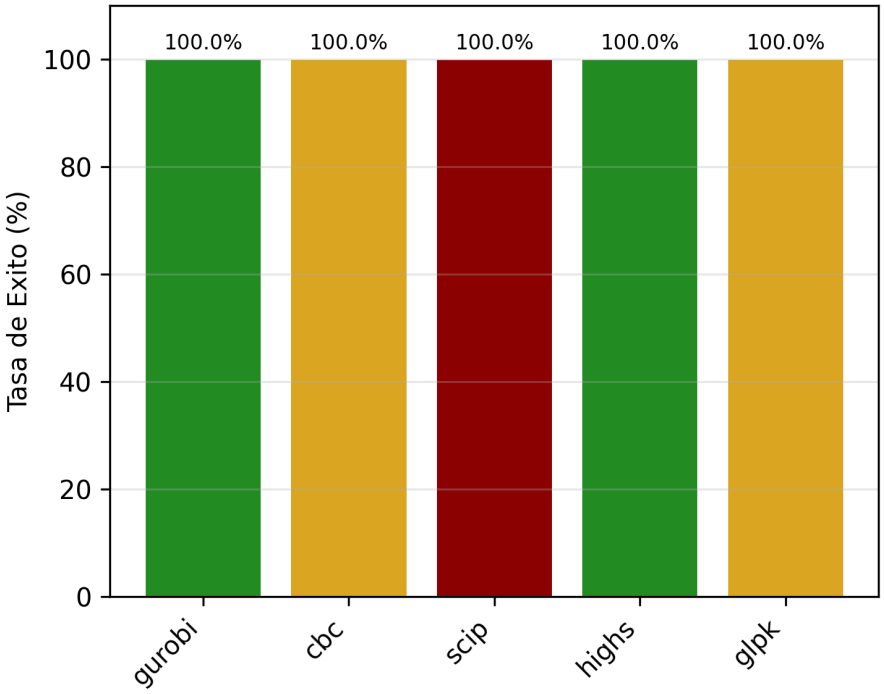
```
MAX Z = 100.0x1 +150.0x2 +120.0x3 +80.0x4 +200.0x5 +90.0x6 +110.0x7
1.0x1 +1.0x2 +1.0x3 +1.0x4 <= 500.0
1.0x3 +1.0x4 +1.0x5 +1.0x6 <= 400.0
1.0x1 +1.0x5 +1.0x6 +1.0x7 <= 300.0
2.0x2 +1.0x3 +1.0x5 <= 350.0
1.0x1 +3.0x2 +2.0x4 +1.0x7 <= 600.0
1.0x3 +1.0x4 +2.0x5 +1.0x6 +1.0x7 <= 450.0
1.0x1 +1.0x2 +1.0x3 +1.0x4 +1.0x5 +1.0x6 +1.0x7 <= 800.0
1.0x1 +1.0x3 +1.0x5 +1.0x7 <= 350.0
x1 >= 0.0
```

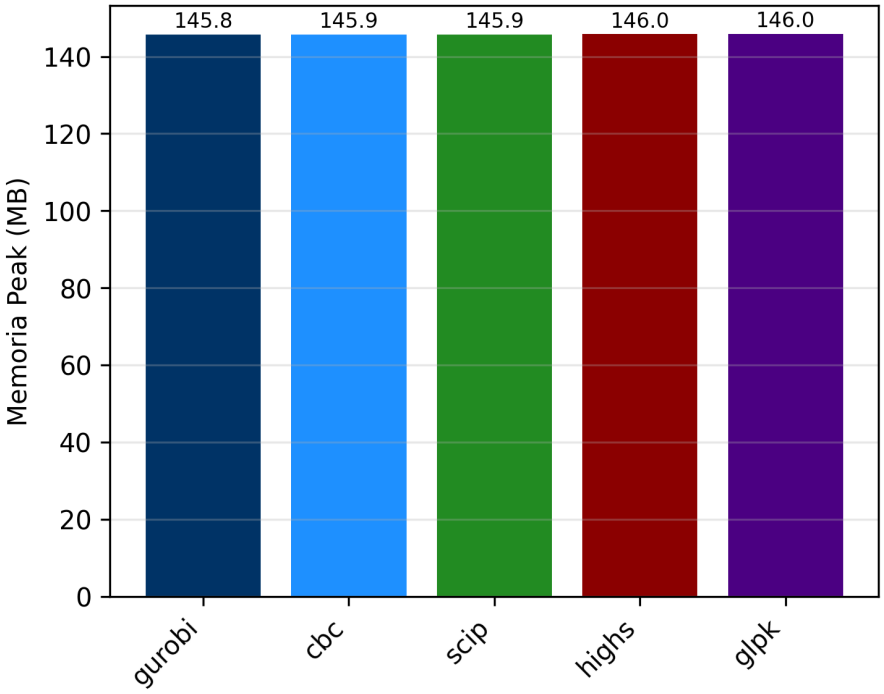
```
x2 >= 0.0
x3 >= 0.0
```

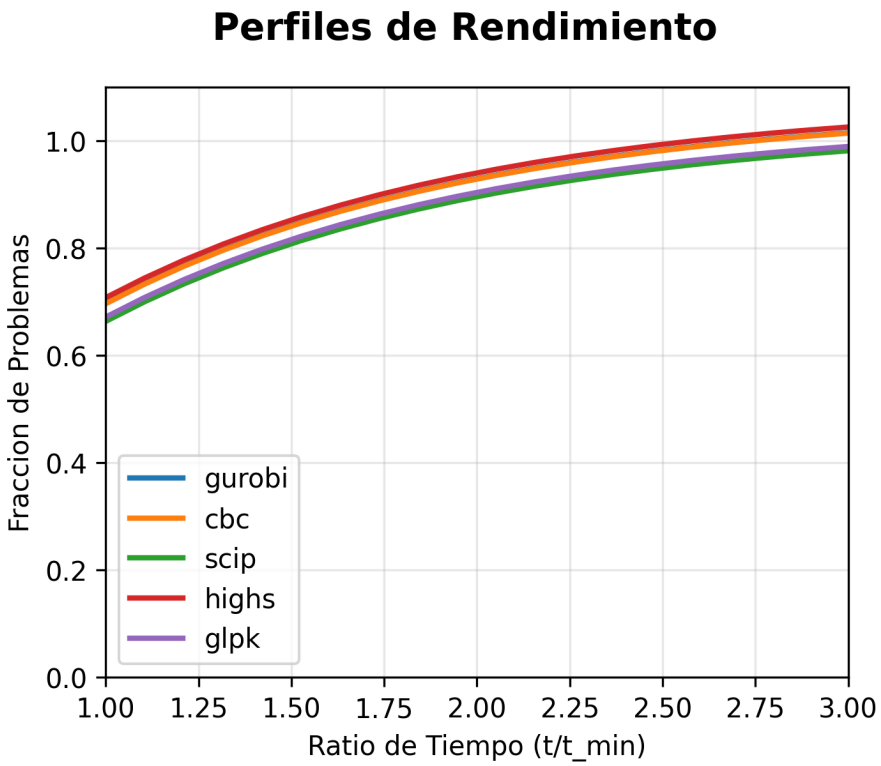
Problema: Problema_10

```
MIN Z = 15.0x1 +20.0x2 +12.0x3 +18.0x4 +25.0x5 +30.0x6
1.0x1 +1.0x2 +1.0x3 >= 100.0
1.0x3 +1.0x4 +1.0x5 >= 150.0
1.0x5 +1.0x6 >= 80.0
2.0x1 +1.0x2 +1.0x4 <= 300.0
1.0x2 +3.0x3 +1.0x5 <= 250.0
1.0x1 +1.0x4 +2.0x5 +1.0x6 <= 280.0
1.0x1 +1.0x2 +1.0x3 +1.0x4 +1.0x5 +1.0x6 = 400.0
2.0x1 +1.0x2 +3.0x3 +1.0x4 +1.0x5 <= 500.0
1.0x1 +2.0x2 +1.0x3 +2.0x4 +3.0x5 +1.0x6 <= 600.0
1.0x1 +1.0x3 +1.0x5 +1.0x6 >= 200.0
1.0x2 +1.0x4 +1.0x5 +1.0x6 >= 180.0
```


Tiempo Promedio por Solver
Tiempo (ms)
200
175
150
125
100
75
50
25
0
129.0
193.8
126.0
124.9
130.0
gurobi
cbc
scip
highs
glpk


100.0%
100.0%
100.0%
100.0%
100.0%
Tasa de Exito (%)
100
80
60
40
20
0
gurobi
cbc
scip
highs
glpk


145.8
145.9
145.9
146.0
146.0
Memoria Peak (MB)
0
20
40
60
80
100
120
140
gurobi
cbc
scip
highs
glpk


Perfiles de Rendimiento
Fraccion de Problemas
Ratio de Tiempo (t/t_min)
0.0
0.2
0.4
0.6
0.8
1.0
1.00
1.25
1.50
1.75
2.00
2.25
2.50
2.75
3.00
gurobi
cbc
scip
highs
glpk

ANALISIS DE ESCALABILIDAD

Relacion entre el tamano del problema (variables/restricciones) y el tiempo de ejecucion.

Problema	Vars	Restr	Tiempo Prom.
Problema_1	2	2	130.7ms
Problema_10	6	11	139.3ms
Problema_2	3	4	129.8ms
Problema_3	4	7	156.6ms
Problema_4	5	6	134.2ms
Problema_5	4	9	138.7ms
Problema_6	6	6	150.2ms
Problema_7	5	7	142.4ms
Problema_8	4	8	148.9ms
Problema_9	7	8	136.5ms

MATRIZ DE CORRELACION

Correlacion entre tiempo, memoria, iteraciones y nodos procesados.

	Tiempo	Memoria	Iter
Tiempo	1.00	-0.04	-0.09
Memoria	-0.04	1.00	-0.03
Iter	-0.09	-0.03	1.00
Nodos	nan	nan	nan

DETECCION DE OUTLIERS

Problemas con tiempos de ejecucion atipicos (outliers estadisticos).

Outliers detectados:

- Problema_3 (166.5ms) - gurobi

- Probler
- Probler
- Probler
- Probler
- Probler
- Probler
- Probler
- Probler
- Probler
- Probler

RECOMENDACIONES

Sugerencias basadas en los resultados del benchmark para seleccionar el solver adecuado.

Ranking de velocidad:

1. highs: 124.9ms prom. | Mem: 0.1MB | Recomendado para velocidad
2. scip: 126.0ms prom. | Mem: 0.3MB | Opcion viable
3. gurobi: 129.0ms prom. | Mem: 0.2MB | Opcion viable
4. glpk: 130.0ms prom. | Mem: 0.0MB | Opcion viable
5. cbc: 193.8ms prom. | Mem: 0.0MB | Opcion viable

Casos de uso:

- Para problemas pequenos (<10 vars): Cualquier solver
- Para problemas grandes: Gurobi (licencia) o HiGHS (open source)
- Para maximizar exito: Verificar formulacion LP valida
- Considerar memoria: Algunos solvers requieren mas RAM

Datos de consumo de memoria:

- gurobi: Prom=0.2MB, Peak=145.8MB
- cbc: Prom=0.0MB, Peak=145.9MB
- scip: Prom=0.3MB, Peak=145.9MB
- highs: Prom=0.1MB, Peak=146.0MB
- glpk: Prom=0.0MB, Peak=146.0MB

INFORMACION DEL SISTEMA

Sistema:	Windows 11
Maquina:	AMD64
Procesador:	Intel64 Family 6 Model 126 Stepping 5, GenuineIntel
Python:	3.14.0
Hostname:	OYHS
Fecha:	2026-05-03T23:16:00

ANALISIS DE MEMORIA

Patrones de uso de memoria y analisis de eficiencia de memoria por solver.

Solver	Mem. Prom.	Mem. Peak	Eficiencia	Problemas
gurobi	0.2 MB	145.8 MB	30.00	30/30
cbc	0.0 MB	145.9 MB	30.00	30/30
scip	0.3 MB	145.9 MB	30.00	30/30
highs	0.1 MB	146.0 MB	30.00	30/30
glpk	0.0 MB	146.0 MB	30.00	30/30

Eficiencia = Problemas resueltos / Memoria promedio. Valores mas altos indican mejor uso de recursos.